10.1: Trails, Paths, and Circuits

Definition.

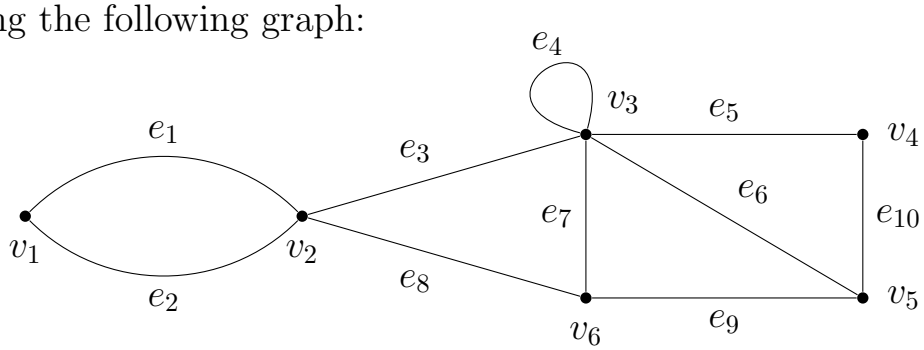
- Let G be a graph, and let v and w be vertices in G.
- A **walk from** v **to** w is a finite alternating sequence of adjacent vertices and edges of G. A walk has the form

$$v_0\ e_1\ v_1\ e_2\ \ldots\ v_{n-1}\ e_n\ v_n,$$

  where the v's represent vertices, the e's represent edges, $v_0 = v$, $v_n = w$, and for each $i = 1, 2, \ldots, n$, v_{i-1} and v_i are the endpoints of e_i. The **trivial walk from** v **to** v consists of the single vertex v.
- A **trail from** v **to** w is a walk from v to w that does not contain a repeated edge.
- A **path from** v **to** w is a trail that does not contain a repeated vertex.
- A **closed walk** is a walk that starts and ends at the same vertex.
- A **circuit** is a closed walk that contains at least one edge and does not contain a repeated edge.
- A **simple circuit** is a circuit that does not have any other repeated vertex except the first and last.

	Repeated Edge?	Repeated Vertex?	Starts and Ends at the Same Point?	Must Contain at Least One Edge?
Walk	allowed	allowed	allowed	no
Trail	no	allowed	allowed	no
Path	no	no	no	no
Closed walk	allowed	allowed	yes	no
Circuit	no	allowed	yes	yes
Simple circuit	no	first and last only	yes	yes

Example. Using the following graph:



determine which of the following walks are trails, paths, circuits, or simple circuits.

$v_1\, e_1\, v_2\, e_3\, e_4\, v_3\, e_5\, v_4$

walk, trail

$e_1\, e_3\, e_5\, e_5\, e_6$

walk

$v_2\, v_3\, v_4\, v_5\, v_3\, v_6\, v_2$

walk, circuit

$v_2\, v_3\, v_4\, v_5\, v_6\, v_2$

walk
circuit
simple circuit

$v_1\, e_1\, v_2\, e_1\, v_1$

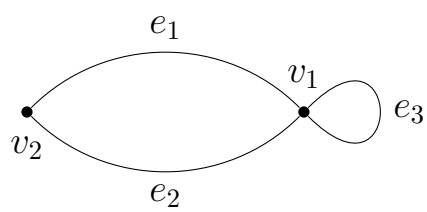
walk

v_1

walk

Definition.
A graph H is said to be a **subgraph** of a graph G if, and only if, every vertex in H is also a vertex in G, every edge in H is also an edge in G, and every edge in H has the same endpoints as it has in G.

Example. Based on the following graph:



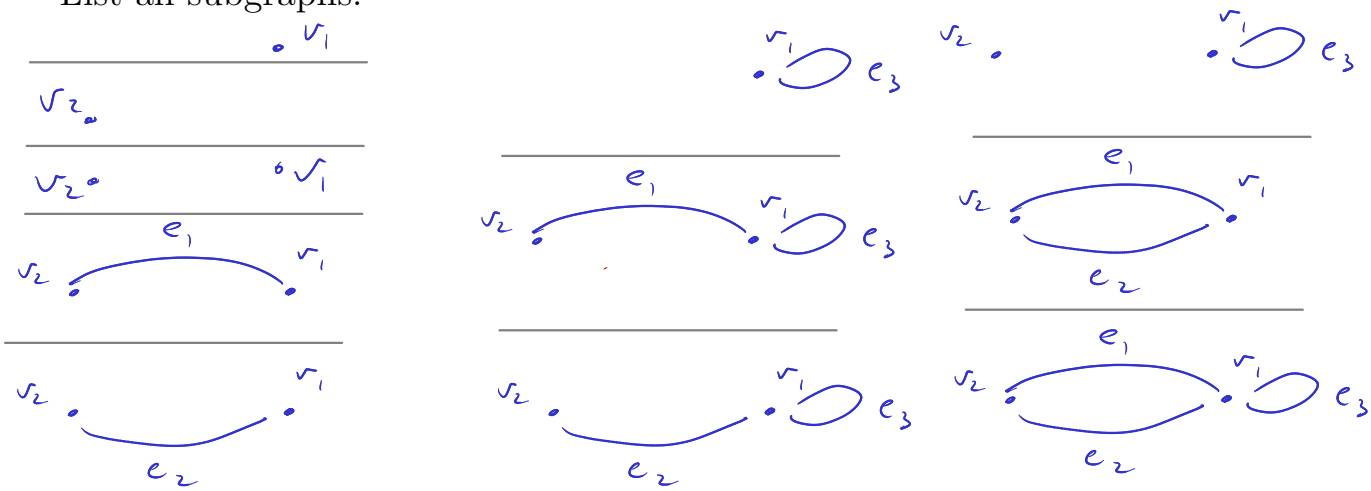
How many paths/trails/walks are there from v_1 to v_2?

Paths
$v_1 e_1 v_2$
$v_1 e_2 v_2$

trails
$v_1 e_1 v_2$
$v_1 e_2 v_2$
$v_1 e_3 v_1 e_1 v_2$
$v_1 e_3 v_1 e_2 v_2$

walks
Infinitely many

List all subgraphs.



Definition.
Let G be a graph. Two **vertices v and w of G are connected** if, and only if, there is a walk from v to w. The **graph G is connected** if, and only if, given *any* two vertices v and w in G, there is a walk from v to w.

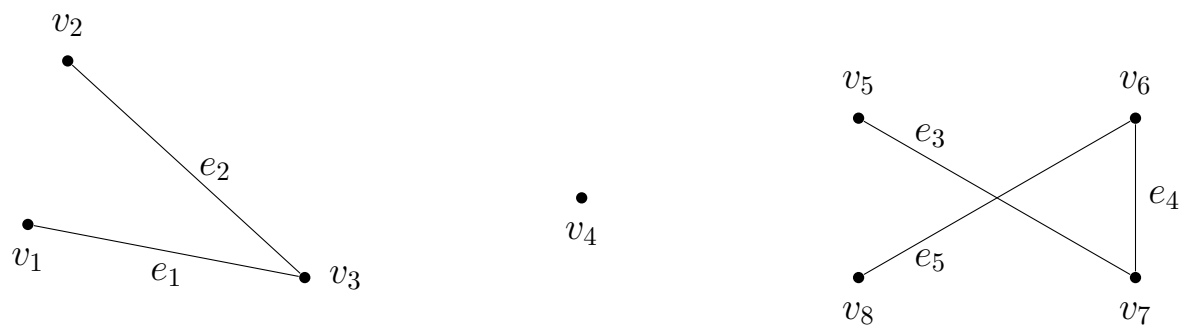
Lemma 10.1.1
Let G be a graph.

1. If G is connected, then any two distinct vertices of G can be connected by a path.
2. If vertices v and w are part of a circuit in G and one edge is removed from the circuit, then there still exists a trail from v to w in G.
3. If G is connected and G contains a circuit, then an edge of the circuit can be removed without disconnecting G.

Definition.
A graph H is a **connected component** of a graph G if, and only if,

1. H is a subgraph of G;
2. H is connected; and
3. no connected subgraph of G has H as a subgraph and contains vertices or edges that are not in H.

Example. Find all connected components of the following graph G



$V_1 = \{v_1, v_3, v_2\}$

$V_2 = \{v_4\}$

$V_3 = \{v_5, v_7, v_6, v_8\}$

$E_1 = \{e_1, e_2\}$

$E_2 = \{\ \}$

$E_3 = \{e_3, e_4, e_5\}$

Definition.
Let G be a graph. An **Euler circuit** for G is a circuit that contains every vertex and every edge of G. That is, an Euler circuit for G is a sequence of adjacent vertices and edges in G that has at least one edge, starts and ends at the same vertex, uses every vertex of G at least once, and uses every edge of G exactly once.

Theorem 10.1.2
If a graph has an Euler circuit, then every vertex of the graph has positive even degree.

Theorem 10.1.3
If a graph G is connected and the degree of every vertex of G is a positive even integer, then G has an Euler circuit.

Theorem 10.1.4
A graph G has an Euler circuit if, and only if, G is connected and every vertex of G has positive even degree.

Definition.
Let G be a graph, and let v and w be two distinct vertices of G. An **Euler trail from v to w** is a sequence of adjacent edges and vertices that starts at v, ends at w, passes through every vertex of G at least once, and traverses every edge of G exactly once.

Definition.
Given a graph G, a **Hamiltonian circuit** for G is a simple circuit that includes every vertex of G. That is, a Hamiltonian circuit for G is a sequence of adjacent vertices and distinct edges in which every vertex of G appears exactly once, except for the first and the last, which are the same.

Proposition 10.1.6
If a graph G has a Hamiltonian circuit, then G has a subgraph H with the following properties:

- H contains every vertex of G.
- H is connected.
- H has the same number of edges as vertices.
- Every vertex of H has degree 2.

Definition.
If G is a simple graph, the **complement of** G, denoted G', is obtained as follows: The vertex set of G' is identical to the vertex set of G. However, two distinct vertices v and w of G' are connected by an edge if, and only if, v and w are not connected by an edge in G.